

LIEW MING KHANG

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SUMMARY

Driven and adaptable engineer passionate about advancing sustainability, artificial intelligence, and robotics. With a foundation in Mechatronics Engineering and strong skills in design and problem-solving, I aim to develop interdisciplinary solutions that tackle real-world challenges and promote a sustainable, technologically advanced future.

WORK EXPERIENCE

Design Engineer Intern

Mar 2024 - Aug 2024

Verdant Solar

- Designed residential solar PV systems using AutoCAD and PVSyst, optimizing layouts for maximum energy production.
- Produced Single Line Diagrams (SLD) and prepared NEM and SEDA documentation, ensuring safe and compliant system designs.
- Streamlined project workflows with Python, enhancing overall efficiency by 25%.
- Assisted in site visits and troubleshooting activities, gaining hands-on experience in solar system operations and maintenance.

PROJECTS

Plastic Sense – Smart Recycling Bin with AI-Powered Plastic Classification

Sep 2024

- Goal: Improve recycling efficiency by automating plastic waste sorting, compacting, and tracking.
- Action: Integrated Time of flight (TOF) sensors for plastic type classification (PET, LDPE, HDPE, Other). Designed automated sorting and high-force compacting mechanisms to streamline processing. Developed a mobile app for real-time tracking of recycling points and user engagement. Implemented deep learning algorithms for AI-powered classification and contamination detection.
- Result: Achieved 75% plastic type detection accuracy and 68% contamination detection accuracy.

ROS2 TurtleBot Navigation and Mapping

Sep 2024

- Goal: Demonstrate autonomous navigation in a supermarket-like environment using SLAM and ROS2 Navigation Stack.
- Action: Designed Gazebo simulation replicating real-world aisles and obstacles; implemented SLAM with lidar and odometry for mapping; developed a PyQt5 GUI to integrate Navigation2 features, manual control, and waypoint management; incorporated error handling, obstacle avoidance, and real-time feedback visualization.
- Result: Achieved accurate 2D mapping with high fidelity to environment layout; robot successfully navigated narrow pathways and avoided static obstacles without collisions in multiple test scenarios, validating robust path planning and autonomous navigation performance.

Swarm Robotics for Adaptive Locomotion and Task Execution on Solar Panel Arrays

Feb 2025

- Goal: Develop a scalable, waterless, and sustainable robotic solution to improve solar panel efficiency by addressing dust and debris accumulation, while reducing maintenance costs and reliance on manual labor.
- Action: Designed and built a robotic prototype integrating locomotion, adhesion, and navigation mechanisms capable of operating on inclined and uneven solar panel surfaces. Implemented decentralized swarm control algorithms using ROS2 and ESP32 microcontrollers for autonomous coordination, edge detection, and obstacle avoidance.
- Result: Delivered a sustainable, modular system design aligned with renewable energy optimization goals and scalable to large solar farms

EDUCATION

Bachelor of Mechatronic Engineering with Honours

May 2021 - May 2025

Asia Pacific University

- GPA: 3.4, Relevant Courses: Mechanical, Electrical, and Software systems

SKILLS

Languages English, Malay, Chinese

Tools/Software SolidWorks, AutoCAD, MATLAB, Ubuntu, Embedded System (Arduino, Raspberry Pi, ESP32)

Programming Languages C++, Python, ROS2

Other Skills Research, Prototyping, Project Management, Documentation, 3D Printing

AWARDS

Can You Hack It 2023 Hackathon

May 2023

- Achieved 3rd place at the Can You Hack It 2023 Hackathon by developing a Generative AI-powered Python application.

Programmable Logic Controller (PLC) Design Competition SEA 2023

Jun 2023

- Designed and implemented ladder logic diagrams to control automated processes.
- Developed a Human Machine Interface (HMI) for system monitoring and user interaction.

Deriv AI Hackathon

Jun 2024

- 2nd place at the Deriv AI Hackathon by developing "Dorin," a Generative AI-powered LLM tool that transforms user insights into actionable steps

REFERENCES

Lennard Leong Jun Shern (Internship Supervisor)

- Senior Design Engineer, +60125867336, lennard@verdantsolar.my

Ir. Eur. Ing. Ts. Dr. Lau Chee Yong (FYP Supervisor)

- Lecturer, +60389961000, dr.laucheeyong@apu.edu.my